

C-HD: Directed Shortest Paths

Informal proof and formalization map - snapshot of September 20, 2026

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PAPER_CHD v2.0: Directed SSSP in $O(n + m + m \cdot \lg(m/n) + m^{1/3}(n \lg n)^{2/3})$ time, formalized for $d \leq \lg^{3/4} n$

Final paper version for /research/final/PAPER.md. Integrated by agent-03.

STATUS (2026-09-21 ~00:00 UTC). The formal statements of §1 (Theorem 1 and Corollary 2) build with the standard axioms only ([propext, Classical.choice, Quot.sound]) in agent-10's frozen integration build #33 (Frontier.CHD.Final). Corollary 2' and the formal space statement are corollaries in files outside that closure (FinalExtras, FinalSpace; std axioms, verified 23:03 UTC). They are claimed as formal only once the post-freeze build #34 that adds them to the root has been audited. - GOAL.md's acceptance pipeline is in progress (§9.2 item 6): kernel replay PASSED (agent-09); fresh-copy rebuild PASSED and REVIEW_1 ACCEPTED (agent-07); REVIEW_2 (agent-02) pending. - This text is final/PAPER.md only once that pipeline passes. Until then it is a candidate. - Theorem 3 (the full density range) is a paper-level statement and is NOT claimed as formalized.

Authors: - agent-09: card C-HD, FindPivots-HD, DS'. - agent-03: integration, §4.3 Lemma L, the §5 time analysis (handled ranges, home charging), §6 parameters, §8 novelty.

Formalization, by lane: - agent-01: Layer A; level body. - agent-02: labels; scans; merge. - agent-03: time-analysis lemmas; tree layer; PT; base case; finalization. - agent-04: D layer. - agent-05: pull. - agent-06: master cost; post-half. - agent-08: spine and loop. - agent-09: FindPivots, B-L2. - agent-10: dispatcher, L6, final assembly. Reviewers: agent-07 (reviewer #1) and agent-02 (reviewer #2).

Inherited background: the B1 draft (/research/background/PRIOR_B1_UNVERIFIED_DRAFT.md; UNVERIFIED; a different team's text), namely S1 (labels, Claim C, L3+), S2 (corrected core, Lemma S2.1) and S5 (data structure D). Everything used is re-verified; the formal chain of §9 does not rely on the draft. - S1 §D and S2 were

checked line by line by agent-02 (REVIEW #2 round 2, board 11:16, no error). - The interface points C1-C5 where C-HD touches the core are handled in §4. Line numbers BM.x, BC.x, MP.x and PT.x refer to B1 pseudocode PCv2.3 (B1 Appendix A). FH.x refers to §3.3 below.

1. Model, problem, theorem

- Input: a directed multigraph $G = (V, E)$, $n = |V|$, $m = |E|$, source s , weights $w: E \rightarrow \mathbb{R}_{\geq 0}$. Parallel edges, self-loops, zero weights, zero cycles and unreachable vertices are allowed.
- Output: for each v , $\text{dis}_G(s, v)$, or $\top$ if v is unreachable. A labeled vector; no order is required.
- Model: comparison-addition on weights (one comparison or addition of reals = 1 step). Unit-cost word RAM on $O(\log n)$ -bit words for indices, counters, pointers and ids. No floor, hashing or bit tricks on reals.
- Deterministic worst case. All input reading, preprocessing and output are charged.

Formal model (Frontier.CostModel, “CostModel v2”). A deep-embedded RAM with a word cap $(n+m+2)^{e+1}$, $e = 160$, and real registers that are only compared and added. - Every statement, test and allocated cell is charged. - The input is read from arrays and the output (reach, dist) is materialized. - Program.Exact means $\text{dist}(v)$ is the infimum over walks, i.e. exact labeled distances. - $\text{RunsWithin } G \text{ s } T$ means the unique terminating run costs at most T .

Theorem 1 (formal: Frontier.CHD.Final.chd_exact_within, chd_CHDTarget). - Let $F(n) = \lfloor \log_2 n \rfloor^{\lfloor 3/4 \rfloor}$, computed as $\text{Nat.sqrt}(\text{Nat.sqrt}(\text{Nat.log } 2 \text{ } n)^3)$. The inner floor matters: e.g. at $\log_2 n = 21.9$, $\lfloor 21^{\lfloor 3/4 \rfloor} \rfloor = 9$ while $\lfloor 21.9^{\lfloor 3/4 \rfloor} \rfloor = 10$. - Let $\text{chdProgram} = \text{chdProg } 160 \text{ psC coreC}$; it is one uniform, closed and deterministic program. - Then chdProgram is exact on every directed graph with nonnegative real weights and every source, and every run costs at most $C \cdot \text{Tdisp}_F(n, m)$, with the explicit constant $C = \text{bodyC KcC} + 65536.9 + 100$. - Here $\text{Tdisp}_F(n, m)$ is: - $\text{Tchd}(n, m) = n + m + m \cdot \ln(m/(n+1) + 2) + m^{\lfloor 1/3 \rfloor \cdot (n \cdot \ln(n+2)) \lfloor 2/3 \rfloor}$, if $m \leq n \cdot F(n)$; - $(n+1)(m+1)$, otherwise (verified Bellman-Ford fallback). - The program’s own dispatcher test runs C-HD when $\log_2 n \geq 16$ and $\lceil m/n \rceil \leq F(n) \approx \lg^{\lfloor 3/4 \rfloor} n$, and Bellman-Ford otherwise. The $C \cdot \text{Tchd}$ bound still holds for every n with $m \leq n \cdot F(n)$: for $\log_2 n < 16$ the Bellman-Ford cost is absorbed by the term 65536.9 (§6).

Corollary 2 (formal: Frontier.CHD.Final.chd_gateC : Frontier.GateC). - Take the profile $\mu(n) = n \cdot \lfloor \ln(n+2)^{\lfloor 3/4 \rfloor} \rfloor$; eventually $n \cdot \sqrt{\ln(n+2)} \leq \mu(n) \leq n \cdot \ln(n+2)$. - Along μ , the bound satisfies $C \cdot \text{Tdisp}_F(n, \mu(n)) / (n \cdot \ln(n+2)) \rightarrow 0$. At $d = \ln^{\lfloor 3/4 \rfloor} n$ it is $O(n \cdot \lg^{\lfloor 11/12 \rfloor} n)$. - In this window the best previously known deterministic directed comparison-addition bound is $\Theta(n \lg n)$. It is the minimum of: - Dijkstra, $O(m + n \lg n)$; - DMM+25, $O(m \lg^{\lfloor 2/3 \rfloor} n)$; - DMSY26, $O(m \sqrt{\lg n} + \sqrt{mn \lg n \lg \lg n})$. - Formal comparison (post-freeze, build #34: agent-01’s Frontier.CHD.FinalCompare.chd_beats_known; std axioms, re-checked by agent-03 at 23:25). The witness bound $T = C \cdot \text{Tdisp}_F$, valid on every graph, satisfies $T(n, \mu(n))/b(n, \mu(n)) \rightarrow 0$ for each of the three bound EXPRESSIONS b , taken with constant 1 and $L = \ln(n+2)$: - $m + n \cdot L$; - $m \cdot L^{\lfloor 2/3 \rfloor}$; - $m \cdot \sqrt{L} + \sqrt{m \cdot n \cdot L \cdot \ln L}$. That these are the best known bounds is the novelty review (§8), not a formal statement.

Corollary 2’ (strong form, all graphs in the range; formal, post-freeze build #34: Frontier.CHD.FinalExtras.chd_auditGateC : Frontier.Audit.GateC Frontier.Audit.ramModel). - Status: FinalExtras is outside the frozen closure of build #33. It compiles with std axioms (agent-10, 23:03) and enters the root at build #34. Corollary 2’ is claimed as formal only once build #34 is audited. - Same witness chd-Program; the implication from CHDTarget F is agent-09’s independent AuditGateC.chdTarget_imp_auditGateC. - Audit’s own specification: Solves means the decoded output equals Audit’s walk-infimum distance adist , and time is the charged cost of the unique terminating run. - Put $f(n) = \max(1, F(n))$. Then $f(n) \geq \text{sqlg}(n) = \max(1, \lfloor \sqrt{\lfloor \log_2 n \rfloor} \rfloor)$. - For every constant K there is N such that EVERY graph with $n \geq N$ and $m \leq n \cdot f(n)$, with every source, has $K \cdot \text{time} \leq n \cdot \lg(n)$, where $\lg(n) = \max(1, \lfloor \log_2 n \rfloor)$. - So the $o(n \lg n)$ bound holds uniformly over all graphs of average degree $\leq f(n)$, not only along the profile μ .

Theorem 3 (paper level, NOT formalized beyond Theorem 1's range). - The C-HD algorithm of §3 runs in $O(T_{chd}(n, m))$ for every m , by §5-§6. - Hence it is $o(n \lg n)$ for every $d = m/n \in [c \cdot \sqrt{\lg n}, o(\lg n / \lg \lg n)]$. - The formalized program switches to Bellman-Ford above $d = F(n)$. The kernel-checked improvement therefore covers $d \leq F(n)$; the range $F(n) < d = o(\lg n / \lg \lg n)$ rests on this paper's analysis only.

Space. - Formally, $\text{space} \leq \text{cost} \leq C \cdot T_{disp_F}$: agent-05's Frontier.CHD.FinalSpace.chd_space_within and agent-01's Frontier.CHD.FinalExtras.chd_space, both from the model invariant RAMAudit.run_space_le_cost. Both are post-freeze files for build #34, as for Corollary 2'. Within frozen #33 the same bound is L6.chdProg_space (NamedWitness), stated for chdProg e ps core under CoreSpec. - The formalized DS' preallocates $\Theta(T_{nat})$ cells (§7), which is superlinear in the Gate-C window. - An $O(n + m)$ -space variant (a reallocating implementation) is sketched in §7 and is NOT formalized.

2. Preprocessing (cost $O(n + m + m \lg \delta)$); this is L6 v2 (agent-10), the canonical layout (COORD G2-5 D6)

- P1. Keep-set, no BFS. $\text{keep}(v) : \Leftrightarrow v = s$ or v has a non-loop in-edge. One $O(n + m)$ pass.
 - A vertex outside keep has no non-loop in-edge, so it is unreachable from s . Its output is $\top$ and its out-edges are dropped.
 - $|\text{keep}| \leq m + 1$. The core's parameters are computed from $(cn, cm) := (\min(n, m+1), m)$.
- P2. Dedup. For kept u and each $v \neq u$, keep the FIRST minimum-weight edge $u \rightarrow v$; self-loops are dropped. The result is SIMPLE and distance-preserving. This is needed by HD1 (§3.3) and by the prefix fact (P).
- P3. Degree reduction by OUT-degree CHAINS. $\delta := \max(3, \lceil 2m/n \rceil)$ is computed from the input (n, m) .
 - A kept u of kept out-degree d_u becomes $c_u = \max(1, \lceil d_u/(\delta-1) \rceil)$ virtual vertices $x = \text{off}[u] + q$, $q < c_u$.
 - A non-last chunk holds exactly $\delta - 1$ real out-edges plus the zero-weight chain edge $x \rightarrow x+1$. The last chunk holds the remaining real edges.
 - Every real edge $u \rightarrow v$ enters $\text{gRep}[v] := \text{off}[v]$, the FIRST chunk of v .
 - Out-degrees are $\leq \delta$. IN-DEGREES ARE UNBOUNDED; $\sum_x \text{indeg}(x) = gM \leq 2m$ (proved: L6/Sizes; the chain edges number $\leq m/(\delta-1)$, so $gM \leq 1.5m$ for $\delta \geq 3$, but only $gM \leq 2m$ is used). $N := gN \leq 2cn$ vertices (L6/Sizes).
 - Every chunk of v has distance $\text{dis}(v)$, and $\text{gRep}[v]$ has the fewest hops.
 - Everything in §5 that uses "degree $\leq \delta$ " uses OUT-degrees only: relaxation targets (B4'), FindPivots scans, base-case inserts.
- P4. Out-lists: a static CSR range $[\text{gSt}[x], \text{gSt}[x+1])$ per virtual vertex x .
 - Each range is sorted by the key $(w(e), \text{head } e)$, chain edge included. Heads in a range are distinct, so the order is strict and agrees with κ for a common prefix label. Cost: $O(\sum \deg \lg \deg) = O(m \lg \delta)$ comparisons (agent-06's verified merge sort).
 - A MUTABLE singly linked live list (head $\text{gHd}[x]$, successor gNxt) initially lists the whole range in order.
 - FH.9 unlinks the slot being scanned through the scanner's predecessor slot, in $O(1)$. This is equivalent to the doubly linked list of v1.2, because FH.9 deletes only the slot under the scan.
- P5. Output. $\text{reach}[v] := \text{gKeep}[v] \cdot \text{gReach}[\text{gRep } v]$ and $\text{dist}[v] := \text{gDist}[\text{gRep } v] = d[\text{gRep } v].\text{len}$. $O(n + m)$.

3. The algorithm

3.1 Labels and order (inherited, B1 S1 §A)

- A label is ∞ or a 5-tuple $(\text{len}, \text{hops}, v, e, a)$: len real, hops int, v the vertex, e the last edge, $a = \text{ver}[\text{tail } e]$ at write time.
- The order $\prec$ realizes the full walk order κ exactly, with one real comparison plus $O(1)$ integer comparisons.
- $\text{Relax}(u, e, B)$: $\text{cand} := d[u] \oplus e = (d[u].\text{len} + w(e), d[u].\text{hops}+1, \text{head } e, e, \text{ver}[u])$.
 - Valid iff $\text{cand} \preceq d[v]$ and $\text{cand} \prec B$.
 - A strict decrease writes $d[v]$ and increments $\text{ver}[v]$; equality is a re-confirmation.
 - Facts used: T1 (monotone), T2 (strict growth), A1 (canonical tree and strict increase along canonical paths), A3 (upper bounds, labels non-increasing), A4 (canonical relaxation).
- PREFIX FACT (P). For fixed $u, e \mapsto d[u] \oplus e$ is strictly increasing along the sorted $\text{Out}(u)$, by T1 and the key order $(w, \text{id head}, \text{id } e)$, which the label compares as $(\text{len}, \text{hops}, v, e)$. So for every bound β , $\{e \in \text{Out}(u) : d[u] \oplus e \prec \beta\}$ is a PREFIX of $\text{Out}(u)$. (agent-02 C5: agent-08’s Lean key must use the same order.)

3.2 The core: B1 PCv2.3 BM.1-BM.31 and BC.1-BC.9 (inherited verbatim), with these changes

- M1 (pointer scans over the STATIC range; two lists, agent-01 11:43, reviewer O8).
 - The main recursion’s scans (BC.7, BM.19-21, BM.27) run over the static CSR range $[gSt[u], gSt[u+1])$, with a pointer $\text{ptr}[u]$ into it. FindPivots-HD (§3.3) scans the mutable live list $gHd/gNxt$.
 - BC.7 (u extracted in a base call with bound B) and BM.27 ($u \in W'$ of a call with bound B): scan the range from its start while $d[u] \oplus e \prec B$, calling $\text{Relax}(u, e, B)$ and inserting as in PCv2.3. Afterwards set $\text{ptr}[u]$ to the first slot with $d[u] \oplus e \succcurlyeq B$.
 - BM.19-21 for $u \in U_i$ (child bound B_i , own bound B): scan from $\text{ptr}[u]$ while $d[u] \oplus e \prec B$, calling Relax and inserting when valid (every scanned e has $d[u] \oplus e \succcurlyeq B_i$, see §4.4). Advance $\text{ptr}[u]$ past the scanned slots.
 - The static range still contains edges deleted by FH.9. By Lemma L(a) they are invalid forever, so relaxing them is a no-op, and the pointer scans each slot at most once per call chain ($O(m)$ extra in total, §4.4).
- M2 (edge deletion). Performed only at FH.9: an edge (u, v) scanned in a FindPivots search with $d[v] \prec L_X$ is unlinked from the live list of u . No other deletion rule is used (agent-02 R2-1: the “flag all in-edges” rule of C_HD_PROOF v1/v2 is dropped).
- M3 (FindPivots-HD) replaces FindPivots (BM.4); §3.3.
- M4 (data structure DS’) replaces S5’s D; §3.4.
- M5 (parameters); §6.
- M6 (BM.22 omitted). The optional pivot switch BM.22 is dropped. S2.5 item 6 shows that removing it keeps every invariant. agent-01’s kernel-checked L4 relation omits it, and agent-07 checked the BM.22-free variant exact on 8000 runs.
- No dispatcher to DMSY26 is used (agent-02 R2-6). The formal dispatcher (Frontier.CHD.Dispatch.dispatchProg) runs the verified Bellman-Ford program (BF.prog, $\text{cost} \leq 9(n+1)(m+1)$) whenever $\log_2 n < 16$ OR $m > n \cdot F(n)$. For $\log_2 n < 16$ this is $O(1)$. Outside the window the Gate-C target makes no claim beyond the fallback bound (§6). ### 3.3 FindPivots-HD (author agent-09; CHD_SEC_FINDPIVOTS.md v1, reproduced with agent-02’s fixes R2-2/R2-3) Search “heap” $H = \text{an UNSORTED ARRAY with a position table: Insert and DecreaseKey } O(1), \text{ ExtractMin } O(|H|)$. $L_X := d_B[S] = \min(\{B\} \cup \{d[x] : x \in S\})$, computed in $O(|S|)$. By Obs. 2.1(2) (S1 §E: the minimum of a frontier is complete), $L_X = \text{dis}_B(S)$.

```

FH.1 compute L_X ; id := new id ; trees := [] ; Wlist := [] ; Q := []
FH.2 for x in S:
FH.3   if fmark[x] = id: continue # x already lies in a tree of this invocation
FH.4   sid := new id ; H := {x} ; K := [x] ; val := {x} ; kst[x] := sid ; kpar[x] := none

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FH.5   while H ≠ ∅ and |K| < k:
FH.6     u := H.ExtractMin()
FH.7     for e = (u,v) in Out(u), in list order:
FH.8       c := d[u] ⊕ e ; if not (c < B): break           # range stop, by (P)
FH.9       if d[v] < L_X: delete e from Out(u) ; continue   # PERMANENT DELETION (Lemma L)
FH.10      if fmark[v] = id:                                # CONTACT with an existing tree (any tree vertex, valid or not)
FH.11        Relax(u, e, B)                                # may improve d[v]; result not needed
FH.12        T := tree(v) ; add K and the edges {kpar[y] : y ∈ K \ {x}} ∪ {(u,e)} to T ; fmark := id on K
FH.13        goto NEXT
FH.14      ok := Relax(u, e, B)
FH.15      if kst[v] ≠ sid:                                  # v new for this search
FH.16        kst[v] := sid ; K.append(v) ; kpar[v] := u      # first-discovery parent (edge e recorded)
FH.17        if ok: val := val ∪ {v} ; H.Insert(v)          # explored member
FH.18        (otherwise v is a LEAF: counted in K, never extracted, label unchanged)
FH.19        if |K| ≥ k: break
FH.20      else if ok:                                       # v ∈ K already and validly (re)relaxed
FH.21        val := val ∪ {v}                                # kpar stays FIRST-DISCOVERY (never re-set)
FH.22        if v ∈ H: H.DecreaseKey(v) else H.Insert(v)    # includes LEAF PROMOTION
FH.23      if |K| ≥ k: create tree T := (K, {kpar[y] : y ∈ K \ {x}}) ; fmark := id on K ; trees.append(T)
FH.24      else: W_x := val ; Wlist.append(W_x) ; Q.append(x) # failed search: H = ∅ and |K| < k
FH.25 NEXT:
FH.26 return MakePivots(S, Q, trees, Wlist, k)             # B1 MP.1-MP.10 (PT pieces of size in [k,3k); empty groups
↪ dropped)

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Local facts (agent-09 §2): - (H-a) heap keys equal current labels; - (H-b) extraction keys strictly increase, and extracted vertices are never validly relaxed again in the same search (T2); - (H-c) K is a tree of graph edges rooted at x, and kpar tails were extracted earlier; - (H-d) the trees of one invocation are pairwise vertex-disjoint, and the FH.12 union is again a tree; - (H-e) $|K| \leq k$.

3.4 Data structure DS' (author agent-09; CHD_SEC_DS.md v1.1). Summary of the parts used below

- Representation. Blocks are unsorted entry lists with disjoint value intervals, in increasing order. They are indexed by a SEPARATOR STACK searched by binary search (v1.5, D2: the code has no balanced tree; BSearch.lean, cost $O(\lg \# \text{blocks})$; the Layer-A model DBlocks charges $\text{bsCost}(\# \text{blocks})$ per search). There is ONE global live pointer $\text{live}[v]$ per vertex; an entry is live iff $\text{live}[\text{key}]$ points to it. $\text{Keys}(D) := \{v : \text{live}[v] \in \text{a block of } D\}$.
- Insert(v,c) in structure D. Let $\text{live}[v] = \text{entry } e0$.
 - If $e0$ lies in D with value $\preceq c$: no change.
 - Otherwise $e0$ becomes stale. This includes an entry in an ANCESTOR's structure (agent-03's rule EA; same effect). A new entry is prepended to the block owning c, found by binary search over the separator stack.
 - (v1.6, agent-02 R-1) Insert NEVER splits (DLazy.insertL; insRAM_spec refines exactly this). Splits are lazy and front-only: they happen inside Pull's front preparation (DLazy.prep; agent-05's prep-Top/splitStep). They are paid by the potential Ψ below, since a full Pull pays M.
 - Cost $O(\lg \# \text{blocks})$.
- Delete: $O(1)$, makes the entry stale.
- Pull = prep, then select (v1.6). prep gathers front blocks, discarding stale entries and splitting the front block lazily, until $M+1$ live entries are held. select picks the M smallest by BFPRT, and the remainder becomes the new first block. PP (S2.0) holds on LIVE entries.
- Merge(D'): splices D' 's blocks into groups of $\geq M/3$ live entries and links them in front; the last small group is spliced onto D's first block. No entry is visited.
 - Cost $O(1 + |D'|/M' + \# \text{groups} \cdot \lg \# \text{blocks}) = O(1 + |D'|/M')$ for $l \geq 2$.
 - (v1.5, D1) There is no base-case BST. A base call runs bounded Dijkstra on an indexed binary heap (IHeap) and, at its return, inserts its leftover heap keys with their labels into a FRESH level-0 DS' structure (BaseDH: newC M_0 B, then one Insert per key). Level 1 then merges an ordinary DS' structure. Each conversion Insert is prepaid by the heap charge ($\text{DC.bins} \geq \text{DCb.bins} + \text{DC.ins } 0 + 3$; BMTeleLoop.baseDH_tele).

- Block count (v1.5, D2: the potential argument Ψ , tracker O29; agent-01 DPsi/BMTele/BMTeleLoop). With $\text{thr } M := \max(1, \lfloor M/3 \rfloor)$, $\text{nsm} := \# \text{blocks with fewer than thr } M \text{ entries}$ and $\Psi(D) := M \cdot \text{nsm} + 2 \cdot \# \text{entries}$:
 - $\# \text{blocks} \leq \text{nsm} + \# \text{entries} / \text{thr } M$ (big blocks hold $\geq \text{thr } M$ entries);
 - a new structure has $\Psi = M$; Insert: $\Psi' \leq \Psi + 2$; a full Pull: $\Psi' + M \leq \Psi$, a short Pull resets $\Psi' = M$; Merge ($M \geq 3$): $\Psi' \leq \Psi + M + 2 \cdot \# \text{entries}(D')$ (grouping leaves at most one small group).
 - Along one call's loop, $\Psi + 2 \cdot (\text{remaining placements of the call's subtree}) \leq 2M_1 + 2E_{\max}(l)$, where $E_{\max}(l)$ is the level-1 placement budget of (B4') (Fits; agent-01 TraceFits from agent-03's ownSum_le/ucap_le). Hence $\# \text{blocks} = O(E_{\max}(l)/M_1 + 1)$ at every Insert, and one Insert costs $O(\lg(E_{\max}(l)/M_1) + 1)$: $\text{insBound}(M, 2M + 2E) \leq 211 \cdot \log_2(8\rho + 8) + 441$ for $E \leq \rho M$ (BMTeleParams.insBound_le). With $M_1 = t \cdot 2^{\lfloor (1-l)t \rfloor}$, $\tau_1 = t^{3 \cdot 2^{\lfloor lt \rfloor}}$, $E_{\max}(l) = (3(L+1) + 3\delta) \cdot 2\tau_1$ this is $O(t)$ in the Lean window.
 - The family potential $\text{potM}(\text{active}) + 2 \cdot \text{stale}(\text{others})$ telescopes along the whole run (bmsspDL_tele_top): the concrete run's cost is at most the cost of the literal run with the amortized charges (pull 735IS'l + 950, delete 5lksl + 1, merge 14, new 3, insert DC.ins l). (v1.4 used “every non-first block had $\geq M/3$ live entries at creation”; that fails for the leftover block of a full Pull and the last group of a Merge, which is why the potential is needed.)
- Semantic equivalence with S5 (agent-09 DS' §4; agent-03 N5.0 “persistence”). During a child's run the ancestors' structures are not read (R9, FP6). Every key inserted inside a child's subtree is, at the child's return, in U_i or in $\text{keys}(D_i)$. So after BM.14-BM.15 the parent's live key set and live values coincide with S5's keep-smaller Merge plus FIX-STALE deletion.
- agent-02 C4: stale entries are invisible to $\text{keys}(D)$, IsEmpty (BM.9, BM.24a, BC.3), PP and BC.8's min.
- (v2.0, agent-04; formal: KeyLists.LL, DRI.kl) IsEmpty in $O(1)$ despite stale entries.
 - Each level keeps a circular doubly-linked list of its live keys.
 - Insert unlinks the key from whatever list holds it; this may be an ancestor's list (rule EA), and unlinking needs no level. Insert then links the key at level l .
 - Delete and Pull unlink. Merge splices the child's list into the parent's in $O(1)$.
 - IsEmpty is $\text{next}[\text{sentinel}] = \text{sentinel}$.
 - Per-level counters would fail: an Insert may supersede an entry whose recorded level is stale after a Merge, and relabelling would cost $\Theta(|D'|)$, which breaks Merge's $O(1 + |D'|/M')$.

4. Correctness

4.1 The inherited core lemma

Lemma S2.1 (B1 S2; checked by agent-02). Every call X satisfying (C1)-(C3) terminates, and its return satisfies (R1)-(R9). This holds for ANY FindPivots routine with FP1, FP2, FP3 (only $W \subseteq \tilde{U}$ is used), FP5, FP6, and ANY D implementing the S2.0 interface (Insert-keep-min, Delete, Merge with the value precondition, Pull with the PP equality and $x \preceq B$, (V1), (V2)). Cor. S2.2: the top call $\text{BMSSP}(\infty, \{x_s\}, L)$ with $\Lambda_L \geq N$ is full, and it makes every reachable vertex complete. C-HD changes only four things: - FindPivots, handled in §4.2 and §4.3; - the scanning order and extent of the core's relaxation loops, in §4.4; - adjacency lists, by deletions, in §4.3; - the implementation of D , in §3.4 / C4. Lemma S2.4 of B1 is FALSE under prefix scans but is not used here (agent-02 C3). S2.5 (one insertion event per edge) uses from it only “all scans of u happen after u is complete”, which still holds.

4.2 FindPivots-HD satisfies FP1, FP2, FP3, FP5, FP6 (agent-09 §4, with agent-02's fixes)

FP1 (frontier). Let $v \in \tilde{U}$. By (C2), $\text{path}(v)$ contains $q \in S$ that is complete at FindPivots' start (and stays complete, A3). - If $q \notin Q$, then q lies in some P_j (FP2), and we are done. - If $q \in Q$, then q 's OWN search ended at FH.24: FH.13 was never taken, FH.19 never fired, and $H = \emptyset$ at the end. (agent-02 R2-2: this holds even if q was later absorbed into another search's tree.) - Let $q = z_0, \dots, z_r = v$ be $\text{path}(v)$. Induction: each z_j is extracted in

q's search with $d[z_j] = \text{dis}(z_j)$. - For the step, take $e^* = (z_j, z_{j+1})$. Then $d[z_j] \oplus e^* = \text{dis}(z_{j+1}) \preceq \text{dis}(v) \prec B$, so e^* lies in the scanned prefix (P). - e^* was never deleted, by Lemma L(b) below: e^* is canonical. - FH.9 does not fire: $\text{dis}(z_{j+1}) \succ \text{dis}(q) \succ L_X$. - FH.10 does not fire. Otherwise q's search would have ended at FH.13. - $\text{Relax}(z_j, e^*, B)$ is valid (A4). So z_{j+1} enters val and H, at FH.17 or FH.22. - By (H-b), z_{j+1} was not extracted before. Since $H = \emptyset$ at the end, it is extracted later with label $\text{dis}(z_{j+1})$. - Hence $v \in W_q$ and v is complete. ■ FP2. $P_1..P_p$ and Q partition S , and each is nonempty. MP assigns each $x \in S \setminus Q$ to the first piece containing it; every such x lies in a tree (FH.3, FH.12 or FH.23); empty groups are dropped (MP.8). Q is excluded from every P_j (B1 c4). FP3. $W = \bigcup_{x \in Q} W_x$, where $W_x = \text{val} \ni x$ has size $\leq |K| < k$. $W \subseteq \tilde{U}$ because val-vertices are x or touched (L3+). FP5 (SP). Every Relax at FH.11, FH.14 or FH.20 has an extracted tail, which is x or was touched earlier. Leaves are never extracted unless promoted by a valid relaxation (FH.22). FP6. No D operation is performed. Lemma F2 (foreign leaves; agent-09). Let v be a tree vertex of X 's FindPivots with $v \notin \tilde{U}_X$. - Then v is a leaf that was never promoted, v is complete, and $L_X \preceq d[v] = \text{dis}(v) \prec B_X$. - Such v is not in S ($S \subseteq \tilde{U}_X$), so it belongs to no P_j (agent-02 R2-3). Proof. It is a leaf because explored vertices are touched and hence lie in $\tilde{U}_X$ (L3+). The range comes from FH.9 not firing and from invalid-and-in-range. Completeness comes from (C1). Promotion would be a valid relaxation into a complete vertex; by A4 that is canonical, so $u = p^*(v) \in \tilde{U}_X$ would force $v \in \tilde{U}_X$. ■

4.3 Lemma L (deletions). Suppose FH.9 deletes $e = (u, v)$ in call X' at time τ . Then:

- (a) for all later labels, $d'[u] \oplus e \succ d'[v]$, so e is never again a valid relaxation;
- (b) $\text{dis}(u) \succ \text{dis}(v)$; in particular e is not the canonical edge of v (A1: $\text{dis}(p^*(v)) \prec \text{dis}(v)$), so no canonical edge is ever deleted;
- (c) no Relax-valid event of the unmodified algorithm is lost. Every insertion event (BM.21, BM.28, BC.7) and every FindPivots relaxation needs a valid relaxation, so the core's executions with and without deletions coincide on all labels, all D operations and all FindPivots outputs. Proof.
 - u was extracted in X' , so $u \in S_{\{X'\}} \cup \text{touched} \subseteq \tilde{U}_{\{X'\}}$ (L3+). Then $\text{path}(u)$ visits some $y \in S_{\{X'\}}$, and $\text{dis}(u) \succ \text{dis}(y) \succ \text{dis}_B(S_{\{X'\}}) = L_{\{X'\}}$.
 - At τ , $\text{dis}(v) \preceq d[v] \prec L_{\{X'\}}$. This gives (b).
 - Later labels satisfy $d'[u] \succ \text{dis}(u) \succ d[v] \succ d'[v]$ (A3), so $d'[u] \oplus e \succ d'[u] \succ d'[v]$ by T2. This gives (a).
 - (c) follows from (a): a Relax call on e would have returned False and changed nothing. This replaces agent-09's open "L_X monotonicity" obligation (agent-02 R2-5): FP1 needs only (b), which is proved directly. ■

4.4 Lemma H0' (pointer scans are faithful; v1.3, two lists)

Setting. $\text{ptr}[u]$ indexes the STATIC CSR range $[gSt[u], gSt[u+1])$, which is sorted by the key of P4. FH.9 unlinks edges only from the mutable FindPivots list $gHd/gNxt$, so it never touches the static range or $\text{ptr}[u]$.

Claim. For every u and every call X on u 's chain $X_0 \supset \dots \supset X_q$ (the calls whose U contains u ; S2.5), consider the moment before X 's BM.19 (or BM.27 / BC.7) scan of u . Then $\text{ptr}[u]$ is the first slot of the static range with $d[u] \oplus e \geq \beta$, where: - $\beta = B_{\{\text{child}\}}$ at BM.19; - $\text{ptr}[u]$ is the range start at BM.27 and BC.7, which is correct since those scan from the start.

Consequences. - The M1 scan calls Relax exactly on the static-range edges e with $d[u] \oplus e \in [\beta, B_X)$. - Edges deleted by FH.9 may still be in that window. By Lemma L(a) they are invalid forever, so Relax on them is a no-op. - PCv2.3's scan of $\text{Out}(u)$ differs only by calling Relax on edges with $\text{cand} \prec \beta$ and with $\text{cand} \succ B_X$. - The first kind are valid re-confirmations or invalid, and never insert, since the BM.20 test fails. - The second kind are invalid. - Hence the executions coincide on labels and D operations. - Each slot is passed by the pointer at most once over u 's chain, so the extra work is $O(m)$ in total.

Proof. - u is complete at every such scan (S2.5), so $c_e := d[u] \oplus e = \text{dis}(u) \oplus e$ is fixed. - (v1.6, agent-02 audit

A-1) In the RAM refinement this is an EXPLICIT premise of winScanD_spec, wScanD_spec and relW_spec: $\forall u \in \text{row}, d u = \text{dis } u \wedge d u \neq \top$. - The spine supplies it from Layer A: for U_i from the sub-call's CallPost; for W' from BM's loop invariant ($W' \subseteq \text{Complete}$). - For the base call it comes from BInv.complete and BInv.UUK (BaseCall0.baseC_fin). - Without it BM.27's pointer claim is false: an $x \rightarrow y$ relaxation inside W' after y 's scan would leave $\text{ptr}[y]$ stale. - The first scan of u is the settling one: BC.7 if u is extracted in a base call, or BM.27 if $u \in W'\{X_q\}$. It scans the prefix $\{c_e \prec B\{X_q\}\}$ from the range start and leaves $\text{ptr}[u]$ at the first $c_e \succcurlyeq B\{X_q\}$. - At $X_{\{r-1\}}$ ($r \leq q$), BM.19 scans from $\text{ptr}[u]$ while $c_e \prec B\{X_{\{r-1\}}\}$ and leaves $\text{ptr}[u]$ at the first $c_e \succcurlyeq B\{X_{\{r-1\}}\}$. - By (P) (the static range is sorted) and the bound chain $B\{X_q\} \preccurlyeq \dots \preccurlyeq B\{X_0\}$ (PP), the pointer invariant holds. - The static range never changes, so there is no deletion case. - Scans in FindPivots (FH.7) use the mutable list and never move $\text{ptr}[u]$. ■ Note: agent-02 C2 lists the three proof steps of S2 that use scans of canonical edges (S2.3 (K3), Step 4 Case 1, Step 5 (R2)). Each canonical edge lies in the relaxed window $[\beta, B_X]$ or in the settling prefix, and by Lemma L(b) it is never deleted from the FindPivots list either.

4.5 Output

By Cor. S2.2 every reachable vertex of the reduced graph is complete at the end. Unreachable vertices keep ∞ . P5 maps labels back as in B1 §8.3. Since the order $\prec$ refines (len, hops), the len-component of a complete label is the real distance. ■(correctness) ## 5. Running time (Lemma 3.9-HD). Author agent-03. Replaces B1 S4.4-S4.6 for C-HD. Parameters (§6): $t \geq t_0$, $k := \lceil \sqrt{t} \rceil$ with $3k \leq t$ (FIX-L13), $M_1 = t \cdot 2^{(1-1)t}$, $\Lambda_1 = t^{3/2} \{t\}$, $\Lambda_0 = t^3$, $L := \lfloor \lg N / t \rfloor + 1$ (v1.5, D6: the code's LF; v1.4 took $\min\{1 : \Lambda_1 \geq N\}$). Then $Lt > \lg N$, so $\Lambda_L > t^3 \cdot N/2 \geq 2048 \cdot N > N$ ($t \geq 16$), and $L \leq \lg N/t + 1$. Degrees are $\leq \delta = \Theta(d)$. $N = O(n)$ is the number of reduced vertices. Size facts inherited from B1 S4.1, with δ in place of 3: - (B1) $|S_X| \leq t^{2/2} \{t\}$ for a level-1 call; - (B2) $\text{partial} \Rightarrow |U_X| > t|S_X|$; - (B3) $|U_X| \leq 2\Lambda_1$; - (B4') [corrected per agent-07 P-1] Under DS', Merge splices blocks of descendants, so D_X 's blocks may hold entries (live or stale) that were inserted anywhere in X 's subtree: relaxation targets of U_X ($\leq \delta|U_X|$, one event per edge), pivots and re-selections. - Every group contains an S-vertex, so $p_Y \leq |S_Y|$, and $\Sigma_{\{Y \subseteq X, \text{ one layer}\}} |S_Y| \leq |U_X|$ by (B2) and full $S \subseteq U$. - Hence $\Sigma_{\{Y \subseteq X\}} p_Y \leq (1+1)|U_X|$. - Re-selections are $\leq \Sigma p + 2\delta|U_X|$ by 5.3: cross and foreign-leaf edges with tail in U_X . - Foreign leaves never enter P, so they need no separate count, and BM.25 re-inserts ($\leq \Sigma_{\{\text{partial } Y \subseteq X\}} |S_Y| \leq |U_X|/t$). Hence $E_{\{D_X\}} = O(|S_X| + (\delta + 1)|U_X|)$, and $\lg(E_{\{D_X\}}/M_1) \leq t + O(\lg t + \lg \delta + \lg L)$. In the Lean-target window ($m \leq n \cdot F(n)$), $\lg L \leq \lg \lg n \leq t$, so Insert costs $O(t)$.

5.1 Handled ranges (N1). Define $B_{\text{low}}(X) :=$ the parent's B' just before X 's iteration (B'_0 from BM.8), and put

$H(X) := [B_{\text{low}}(X), B'_X]$. (a) $H(\text{child}) \subseteq H(\text{parent})$. (b) Siblings' H are disjoint and ordered. (c) Two calls of one layer have disjoint H . (d) $U_X \subseteq \text{dis}^{\wedge}\{-1\}(H(X))$; a FULL call has $H(X) = [B_{\text{low}}(X), B_X]$. Proof. - Monotonicity (J5) and BM.24a give the upper end. - For the lower end, $B'_0(Z) \succcurlyeq B_{\text{low}}(Z)$: (S-a) at Z 's parent gives $\text{dis}(S_Z) \succcurlyeq B_{\text{low}}(Z)$, and BM.8 takes a min over pivots of S_Z (and $B_Z \succ d[y] \succcurlyeq \text{dis}(y)$ for $y \in S_Z \neq \emptyset$). - (d) follows from (R1) and (S-a). ■ Also $L_X = \text{dis}_B(S_X) \succcurlyeq B_{\text{low}}(X)$ by (S-a), so foreign leaves (F2) satisfy $\text{dis}(v) \in [L_X, B_X] \subseteq [B_{\text{low}}(X), B_X]$.

5.2 Pivot bound with foreign leaves (N3)

- Partial X : $p_X \leq |S_X| < |U_X|/t$.
- FULL X : the trees of X are vertex-disjoint (H-d); their non-foreign vertices lie in $\tilde{U}_X = U_X$; the foreign ones form O_X . So $\Sigma_T |V(T)| \leq |U_X| + |O_X|$, and $p_X \leq \Sigma_T (|V(T)| - 1)/(k - 1)$ (PT pieces, MP.8).
- For full $X \neq X'$ of one layer, $U_X \cap U_{X'} = \emptyset$ (5.1(c,d)). Also $O_X \cap O_{X'} = \emptyset$: a foreign leaf is complete, so its dis is fixed, and it lies in both $H(X)$ and $H(X')$, which are disjoint.
- Hence $\Sigma_{\{\text{full } X \text{ in a layer}\}} (|U_X| + |O_X|) \leq 2N$ and $\Sigma_{\{\text{full } X \text{ in a layer}\}} p_X \leq 2N/(k - 1)$.

5.3 Re-selection charging with foreign leaves (N4; replaces B1 T5 “ $F_j \subseteq U_X$ ”, which fails under contact-at-leaf)

Setting: X full, children $Y_1..Y_f$. - Homes: $h(u) = r$ if $u \in U_{\{Y_r\}}$; $h(u) = 0$ if $u \in W'_X$. - A foreign $v \in O_X$ gets $h(v) = r$ if $\text{dis}(v) \in H(Y_r)$; $h(v) = -1$ if $\text{dis}(v) \prec B^*0$; $h(v) = f+1$ if $\text{dis}(v) \succ B^*f$. (i) g_j (the number of BM.23 re-selections of group j) $\leq 1 + c_j$, where $c_j = \#\text{edges of the PT piece } F_j \text{ whose endpoints have different homes}$. Each re-selection at iteration r removes a pivot of home r (BM.18), and there is at most one per (j, r) . Removing the c_j edges leaves $c_j + 1$ single-home components. (ii) Every tree edge (u, e) has an extracted tail $u \in U_X$. - Different-home edges with both ends in U_X lie in $E_X := E(U_X) \cup_r E(U_{\{Y_r\}})$, and $\Sigma_X |E_X| \leq m$ (B1 G3). - A different-home edge (u, v) with $v \in O_X$ can occur at ONE full call only. At a proper full descendant $X' \subset Y_r$ containing u in U : $v \in O\{X'\}$ gives $\text{dis}(v) \in H(X') \subseteq H(Y_r) = \text{home of } u \text{ at } X$, so the edge would not be different-home at X . Hence $\Sigma\{\text{full } X\} \Sigma_j g_j \leq \Sigma_X p_X + 2m$. Partial X : at most $|S_X|$ re-selections (B1 T5 partial).

5.4 Direct cost of one call X at level $l \geq 1$ (children excluded). Constant C' independent of n, m, δ, t .

T1 FindPivots-HD (FH.1-FH.26), by HD1: - L_X , the FH.2 loop and MakePivots/PT: $O(|S_X| + \Sigma_T |V(T)|)$. - Successful and contact searches: $O(k \cdot (\Sigma_T |V(T)| + |S_X|))$, by HD1 classes (c),(d),(e) with $|K| \leq k$. [Each tree or contact search adds $|K|$ new tree vertices and costs $O(|K| \cdot k)$.] - Failed searches: $O(k^2)$ each, i.e. $O(k^2 |Q_X|)$. - Deletions (FH.9): $O(1)$ each, $O(m)$ globally. With 5.2: full X gives $O(k(|U_X| + |O_X|) + k^2 |Q_X|)$. Partial X gives $\Sigma_T |V(T)| \leq k|S_X|$, hence $O(k^2 |S_X|) = O(|U_X|)$ by (B2) and $k^2 \leq 2t$. T1' BM.6-BM.7: $O(|P|) + p_X \cdot O(t + \lg \delta)$. T2 Picking (BM.11-BM.12): $O(k)$ per pulled current pivot. - Partial child: $O(k|S_{il}|) \leq O(|U_{il}|)$. - Full child: absorbed by the $O(t)$ re-selection triggered at BM.18/BM.23, since $k \leq t$. - Groups emptied: $O(kp_X)$ once. (B1 T2.) T3 Pull (BM.10): $O(|S_{il}|) = O(|U_{il}|)$ (full child: $S_i \subseteq U_i$; partial: (B2)), plus stale discards (charged to their creating Insert/Delete). Merge (BM.14, DS'): $O(1 + |D_{il}|/M_{\{1-1\}})$. For $l = 1$ the child's structure is the level-0 DS' structure built at the base return (v1.5, D1), so Merge is the same block-linking Merge; the conversion Inserts were prepaid in the base call. BM.15 deletions, BM.16-18 and BM.30 membership updates: $O(|U_{il}|) + O(|P|)$. BM.19 pointer scans: $O(|U_{il}|) + \#\text{edges relaxed (4.4)}$. BM.24a and BM.26: $O(1 + |W|)$. BM.27 scans: $O(1 + \deg(u))$ per $u \in W'_X$. Each vertex is in W' of at most one call ($W'_X \subseteq U_X$ and $W'_X \cap U_{\{Y_r\}} = \emptyset$ for every child), so this is $O(m + N)$ globally. (agent-02 R2-8: not charged to the failed search.) T4/T7 Insertion events (BM.21, BM.28): $O(t + \lg \delta)$ each (5 intro, (B4)); at most one per edge globally (S2.5 = B1 G2). T5 Re-selections (BM.23): $O(k + t + \lg \delta) = O(t)$ each; counted in 5.3. T6 BM.25: $O((t + \lg \delta)|S_X|)$. Nonzero only for partial X , so $O(|U_X|)$ by (B2), under the SIDE CONDITION $\lg \delta \leq t$ (agent-07 P-3). - Lean-target window: $\lg \delta = O(\lg n) \leq t$. - Case $t = t_0$ of §6: $\Sigma\{\text{partial}\} (t_0 + \lg \delta)|S| \leq (N L/t_0) \cdot O(\lg \delta)$, with $L \leq \lg N/t_0 + 1$ and $N \lg N \leq t_0^{3/2} m$, which is $O(m \lg \delta)$. T0 Overhead $O(1) \leq O(|U_X|)$, since $U_X \neq \emptyset$ (S2 Step 5). Base call ($l = 0$): bounded Dijkstra on an indexed binary heap (IHeap, agent-06; v1.5, D1) over $\leq |S| + \delta \Lambda_0$ entries, $O(\lg(t\delta))$ per operation, followed by one prepaid DS' Insert per leftover key into a fresh level-0 structure (§3.4). - Each vertex is extracted once globally (it is then in U), and its prefix is scanned once (BC.7). - Each edge has at most one BC.7 insertion event (G2). - (v1.6, agent-02 R-2) With the D1 prepayment $DC.\text{bins} \geq DCb.\text{bins} + DC.\text{ins } 0 + 3$, every heap insertion is charged $O(\lg(t\delta)) + DC.\text{ins } 0 = O(t + \lg \delta)$, which includes its later conversion Insert. - The conversions number at most $\Sigma_{\text{base}} (|S| + \#\text{BC.7 insertion events}) \leq N + m$. This uses $\Sigma_{\text{base}} |S| \leq N$, from (B2) and fullness, and (G2). - Base-call total: $O((N + m)(t + \lg \delta))$. - It enters §5.6 only through the $m \cdot t$ and $m \cdot \lg(t\delta)$ terms, and it needs $N = O(m + 1)$. That comes from preprocessing: $|keep| \leq m + 1$ (P1) and $N = gN \leq 2cn \leq 2(m + 1)$ (P3; L6/Sizes). So $N \cdot t \leq 2(m + 1) \cdot t$ is within the $m \cdot t$ term of §5.6.

5.5 Global facts

- (G1) Per layer $\Sigma_X |U_X| \leq N$ (5.1(c,d)); there are $L+1$ layers, and $\#\text{calls} \leq (L+1)N$.
- (G2) $\Sigma \#\text{insertion events} \leq m$.

- (G3) $\sum_X |E_X| \leq m$.
- (G4, IN-DEGREE-FREE; v1.3) $\sum_X |Q_X| \leq gM + 1 + N \cdot L/t$. In-degrees are unbounded after P3, so B1's $N(\delta+1)$ form is not used.
 - Full calls: each Q-appearance of v in a full call X gets the WITNESS wit X v , the in-edge of the latest INSERTING relaxation event into v (BM.21, BM.28, BC.7) before X starts (reviewer O22(a); the predecessor edge of $d[v]$ is NOT injective). Distinct appearances get distinct events, and every edge has at most one inserting event (G2), so $\sum_{\text{full}} |Q_X| \leq \sum_v \text{indeg}(v) + 1 = gM + 1$. The +1 is the source s , which may lie in Q of one full call with no inserting event (O22(b); CostAggregate's src exemption).
 - Partial calls: $|Q_X| \leq |S_X| < |U_X|/t$, so $\sum_{\text{partial}} |Q_X| \leq N \cdot L/t$.
 - FindPivots-HD writes labels but performs no D operation. This includes the FH.11 contact relaxations, which therefore create no Q-charge (agent-07 minor).
 - A Q-root is never inserted into D by FindPivots.
- (G5) $\sum_X |S_X| \leq (L+1)N$: full $S \subseteq U$, partial $|S| < |U|/t$.
- (G6) $\sum$ over layers of full-call foreign leaves $\leq (L+1)N$; $\sum_X p_X \leq 2(L+1)N/(k-1) + \sum_{\{\text{partial}\}} |S_X|$ (5.2).
- (G7) Merge total $O(\#calls + m + \sum |S| + \sum_{\{l \geq 2\}} \sum_{\{\text{partial } Y \text{ at level } l-1\}} |D_Y|/M_{\{l-1\}})$.
 - [agent-07 P-2, block-walk accounting] Merge at a level- l call walks the child's blocks and absorbs every walked block into a group, so each walked block is DESTROYED. Total walk work = $O(\#blocks \text{ ever created}) = O(\#splits + \#Pull \text{ remainders} + \#groups)$.
 - * (v1.6, agent-02 R-1/R-3) Splits happen only in Pull's lazy front preparation. They are paid by the potential Ψ of §3.4, since a full Pull pays M . The eager-split count “ $\#splits \leq \sum \#inserts/(M/2)$ ” of v1.5 described a structure that is not implemented.
 - * $\#Pull \text{ remainders} \leq \#pulls \leq \#calls$.
 - * $\#groups \leq \#merges + \sum_l (\text{live entries bubbled into level } l)/(M_l/3)$.
 - * Live bubbled entries per layer $\leq \sum_{\{\text{partial } Y\}} (|S_Y| + \delta |U_Y|) \leq N + \delta N$.
 - * Spliced groups go on the separator stack at $O(1)$ each; there is no tree (v1.6, R-3).
 - * The formal route does not use this block count: it uses the family potential and $bmsspDL_tele_top$ (§3.4).
 - * Base conversions (the leftover Inserts at base returns, prepaid by the heap charge) are $O(\sum_{\{\text{base}\}} (|S| + \delta |U|)) = O(\delta N + N)$.
 - Merge total: $O(\#calls + \#inserts + m + \delta N/t) = O((L+1)N + m)$.

5.6 Summation

$T_{\text{core}} = O(\sum_X [k(|U_X| + |O_X|) + (t/k)(|U_X| + |O_X|) + |U_X|])$ (T0-T3, T1', T6 over all calls; 5.2 for p) + $k^2 \sum |Q_X|$ (failed searches) + $(t + \lg \delta) \cdot m$ (G2: insertion events) + $t \cdot (\sum p + m)$ (T5 via 5.3; the p term is already inside the first line) + $(N + m) \cdot \lg(t\delta)$ (base calls) + $(L+1)N + m$ (Merge, deletions, BM.27) = $O((L+1) \cdot N \cdot (k + t/k) + k^2(m + 1 + N L/t) + m(t + \lg(t\delta)))$ [(G4) in-degree-free] With $k = \lceil \sqrt{t} \rceil$ ($k^2 \leq 2t$, $t/k \leq \sqrt{t}$): $T_{\text{core}} = O(N \cdot (\lg N/t + 1) \cdot \sqrt{t} + m \cdot t + m \cdot \lg(t\delta)) = O(N \lg N / \sqrt{t} + m t + m \lg(t\delta) + N \sqrt{t})$. ### 5.7 Formal counterparts (v1.3; the Lean names of the counters of CostAggregate.CallCounters) - Fo $X := tv_X \ U_X$, the FindPivots tree vertices outside the returned set (Frontier/CHD/FPForeign.lean, foOf). - Foreign leaves of a full call lie in $H(X) = [B_{\text{low}}(X), B_X]: \text{foOf_range}$, from agent-05's F2 (fpC_foreign). - Valid.foreign_level (counters_foreign_level), Valid.full_p (counters_full_p) and Valid.partial_Fo (counters_partial_Fo) are proved from it. - Homes (Frontier/Homes.lean, Density.Ranges.home). At a call X , $v \in U_Y$ for a child Y is homed at Y . A $v \notin U_X$ is homed at the child whose handled range contains $\text{dis}(v)$. Otherwise v is unhomed. - home_of_mem and home_of_range give uniqueness via sibling disjointness. - cross_of_home_ne: different homes with both ends returned means no child contains both. - split_of_home_ne: different homes with the head foreign means the tail SPLITS $\text{dis}(\text{head})$ at X . - Cr $X / \text{Be } X$ (Frontier/CHD/CrBe.lean, crOf/beOf) are the graph edges of the

different-home tree edges of a FULL call, each tree edge represented by a graph edge whose source (the extracted end) is in U_X . Cr X has head in U_X ; Be X has head a foreign leaf. - Valid.cr_cross: crOf_cross. - Valid.be_total: beOf_total, via Density.Ranges.beta_sum_le. - 5.3(i) $g_j \leq 1 + c_j$, combinatorial core (Frontier/CHD/Reselect.lean): - colors_le_measure: in a rooted list with a strictly increasing parent measure, #colours $\leq 1 + \#bichromatic$ parent edges; - pieces_colors_le: the same on PT pieces; - groups_colors_le: summed over MakePivots groups, $\leq \#groups + \#bichromatic$ tree edges; - bich_le_crbe: bichromatic tree edges of a full call $\leq |Cr X| + |Be X|$. - O10, Valid.cost_le (Frontier/CHD/CostLe.lean, tracedCounters_cost_le). tracedCounters is agent-01's Log.countersC (BMTrace) with Fo = foOf, Cr = crOf, Be = beOf and Del = delOf. - Per-record budgets come from callC_cost/bmsspC_reccost (CostLog). The record facts come from bmsspC_recall (RecF-PLog). - The markings bound is 5.3(i) (mkOf_full_le) for full calls and mkOf_le_S for partial calls. - $|W| \leq k|Q|$ is fpC_W_card (FPWQ). $\#tv \leq |U| + |Fol|$ holds because the forest is nodup. - The arithmetic is budget_arith. - Transfers for foreign_level, full_p, partial_Fo, cr_cross, be_total and del_disj are also proved. - O22, the Q-witness (Frontier/CHD/QWit.lean, QWitProv.lean, QWitValid.lean). - evOf r are the inserting-capable events: J; W' edges below B; base edges out of U below B. - ev_unique (G2): an edge is an event of at most one record, since the event windows along the chain of calls containing its tail are disjoint. - Events are anchored (a jump at the returning child, an own event at the record); Before means "ended before the call started". - LocalProv: a frontier member of a sub-call is in the parent's S Q, or is the head ($\neq s$) of an event anchored in an earlier child subtree, or is in S of a call of such a subtree. It is proved for every BMSSPC derivation (bmsspC_prov) by a key-provenance invariant of the loop store. Two facts drive it: a valid relaxation never inserts s, and eval $\leq \text{ext} \circ d$ under WalkInv. - qwit_of_prov / bmsspC_qwit: wit X v is the edge of an event anchored strictly inside the nearest full proper ancestor with $v \in Q$, which ended before X started. This gives the Valid fields wit_mem, wit_inj (by G2) and src_once (CostAggregate.Valid). QWitValid restates them for tracedCounters. - O12, (B4') Log side (Frontier/CHD/TraceSize.lean, wB_sum_le). Σ over the log of the own-placement weights ($3|S| + |J| + |Wrl|$ for recursive records, $|S| + |Eout U|$ for base records) is $\leq (3(L0+1) + 3\delta) \cdot |U_{\text{root}}|$, given $|S| \leq |U|$. - PT at RAM level (Frontier/CHD/PartitionRAM.lean ... PartitionRAM5.lean, ptProg_spec). - (v1.5) The FindPivots tree layer at RAM level (B-L2; agent-03, instantiating agent-09's interface TreeI): - ForestRep / FR0 (TreeRAM, TreeRAM7): the forest as linked chains (tr.hd/tr.nx/tr.tl/tr.ln, tree ids, parents fp.par); - FH.23 new tree: newTree_spec (TreeRAM2), cost $9|K| + 20$; - FH.12 contact merge: mergeProg_spec (TreeRAM3-6) refines agent-05's mergeTree - path walk along kpath with re-rooting, rest pass K path, tail/length update, path clear - cost $33|K| + 40$; - treeImpl : TreeI (TreeRAM7): grow = if contact then merge else new tree, cost $\leq 41(|K| + 1)$; every Layer-A fact the RAM code needs is derived from agent-05's Search.treeInv, Search.result_info, kpath_spec, growForest_inv; - FP-TAIL (TreeRAM8, TreeRAM9): compaction of the chains into the PT layout (compact_forestAt), S/Q bitmaps, ptProg, cleanup of fp.fm; fpTail_spec stores exactly forestGroups S Q k trees (re-labelled by Fin.val; PartitionMap: Algorithm 5, MakePivots and ParentFirst commute with injective relabellings) at cost $\leq 188\Sigma|T| + 8|S| + 8|Q| + 30 = O(\#tv + |S| + |Q|)$. - WFrame: a syntactic frame lemma (word-only statements change only the registers/arrays they name), used for every Unchanged footprint of the tree layer. - (v1.5) CostLe2.fpA_le_of_hext: with $\text{hext} \leq c_2(k + 1)$ (unsorted-array extract-min), $\text{fpA} \leq (2(\text{scanC} + \text{hins}) + c_2 + 2)(k + 1)$, so the constant of Valid.cost_le is independent of k (reviewer OBJECTION 4).

6. Parameters and the final bound (author agent-03)

- Constants: $t_0 := 16$. Then $k := \lceil \sqrt{t} \rceil$ satisfies $3k \leq t$ and $k \geq 4$ for $t \geq t_0$.
- Small inputs (v1.5, D4). For $\lg n < 16$, and outside the density window ($m > n \cdot F(n)$), the program runs the verified Bellman-Ford fallback (Frontier.CHD.Dispatch, BF.prog, cost $\leq 9(n+1)(m+1)$); for $\lg n < 16$ this is $\leq 9 \cdot 2^{16} \cdot (m+1) = O(m + 1)$. There is no Dijkstra fallback.
- $t := tF(\text{cn}, \text{cm}) = \max(16, \text{tpar}(\text{cn}, \text{cm}))$ with $\lg N^2 \leq t^3 \cdot \text{dd}^2$ (CostSkeleton.tF_spec; dd the density), i.e. $t \approx (\lg N/d)^{2/3} = (N \lg N/m)^{2/3}$ up to rounding; $k := \lceil \sqrt{t} \rceil$ (kF); $L := \lfloor \lg N/t \rfloor + 1$ (LF; v1.5, D6). These are computed by the verified parameter program (L6/Params paramsProg, L6/LevelTab levelTab: cp.tau[1] = $t^3 2^{\lfloor \lg t \rfloor}$, cp.M[l] = M_l exactly) in $O(t + L + \lg N)$ word operations.

(A_M) caveat for DS'. Lemma DS' gives Pull $O(|S'|)$ and split $O(1)$ amortized when $\lg \#blocks = O(M_1)$. In general Pull costs $O(|S'| + \lg B_X)$ and a split $O(M_1 + \lg B_X)$, where $B_X \leq 3N_X/M_1 + 1$ and $\lg B_X = t + O(\lg(t\delta))$. (A_M) fails only in layers l with $t \cdot 2^{\lceil (1-l)t \rceil} < c \cdot (t + \lg(t\delta))$. There are at most $1 + \lceil \lg \lg(t\delta)/t \rceil + 1$ of them. Each such layer has $\leq N$ pulls (each pull feeds a child with nonempty U , and a layer's U 's are disjoint). So the extra is $O(N \cdot (t + \lg(t\delta)) \cdot (2 + \lg \lg(t\delta)/t)) = O(N t + m \lg(t\delta))$. For the last step: after P3, $m \geq N\delta/4$. Also $\lg \lg(t\delta) \leq \delta t$, so $N \cdot \lg(t\delta) \cdot \lg \lg(t\delta)/t \leq 4 \cdot m \cdot \lg(t\delta)$.

Final bound. §2 costs $O(n + m + m \lg \delta)$. §5.6 with the (A_M) extra, and P5 output $O(n + m)$, give $T = O(n + m + m \cdot \lg \delta + N \cdot \lg N / \sqrt{t} + m \cdot t + m \cdot \lg(t))$. - Case $t = \lceil (N \lg N/m)^{2/3} \rceil \geq t_0$. Then $N \lg N / \sqrt{t} = O(m^{1/3} (N \lg N)^{2/3})$ and $m t = O(m^{1/3} (N \lg N)^{2/3})$. Put $x := N \lg N/m \geq 1$. Then $m \lg t = O(m \lg(2 + x))$ and $m \cdot x^{2/3} = m^{1/3} (N \lg N)^{2/3}$. Since $\lg(2 + x) = O(x^{2/3})$, $m \lg t$ is $O(m^{1/3} (N \lg N)^{2/3})$. - Case $t = t_0$. Then $m \geq N \lg N / t_0^{3/2}$, so $N \lg N / \sqrt{t} = O(m)$ and $T = O(m \lg \delta + m)$. - With $N = O(n)$, $\delta = O(m/n + 1)$ and $\lg \delta = O(\lg(2 + m/(n+1)))$, this gives $T = O(\text{TCHD}(n, m))$ for EVERY (n, m) , where $\text{TCHD}(n, m) := n + m + m \cdot \lg(2 + m/(n+1)) + m^{1/3} (n \lg(n+2))^{2/3}$. ■

Lean target form (what the formal proof uses). The frozen statement is CHDTarget F (agent-10's Frontier/GateC.lean), with $F(n) = \lfloor \log_2 n \rfloor^{3/4}$. - Dispatcher (Dispatch.dispatchProg, dispatch_runs). It runs verified Bellman-Ford (cost $\leq 9(n+1)(m+1)$) when $\log_2 n < 16$ or $\lceil m/n \rceil > F(n)$, and C-HD otherwise. For $\log_2 n < 16$ and $m \leq n \cdot F(n)$, the Bellman-Ford cost is $\leq 9 \cdot 2^{16} \cdot (m+1) \leq 9 \cdot 2^{16} \cdot \text{Tchd}$; this is the term 65536.9 of C. - On the C-HD branch the core gets the density bound $\text{dd}(cn, cm) \leq F(cn) + 1$ (L6 CoreIn.dens). It is used in exactly one place, agent-06's master cost (Frontier/CHD/MasterCost.lean, Part 2). - There the recursive DS' insertion charge satisfies $211 \cdot \log_2(8\rho+8) + 441 \leq 3826 \cdot t$ (mc_ins_le), via $\lg N \leq 16 \cdot t^6$ (mc_lgN_le). - Here $\rho = (3(L_0+1) + 3\delta) \cdot 2t^2 \cdot 2^t$, and $E_{\max_l} \leq \rho \cdot M_1$ at every recursive level $l \geq 1$, where $E_{\max_l}$ is the level- l entry bound (BMTeleChd.chdE $_{\max_l}$). So the $\log_2(\#blocks)$ part of an insertion is $O(t)$ on the branch. - This replaces the (A_M) case analysis above in the formal proof. The level-0 conversion charge needs no density bound (mc_ins0_le). - The other parameter facts hold for every input, with $t = tF = \max(16, \text{tpar})$: $-k(k+1) \leq 2t$ (mc_kk); $-\lg N + 1 \leq LF \cdot t$ (mc_lgN_lt); $-\tau(LF) > 2n$ (mc_tau_top); $-\lg N^2 \leq t^3 \cdot \text{dd}^2$ ($\text{CostSkeleton.tF_spec}$). - So the formal proof has no separate case $t = t_0$ and no threshold n_0 : the bound C.Tchd holds on every input of the branch, with the explicit C. Also $T_{\text{nat}} \leq 33 \cdot \text{Tchd}$ for every $n \geq 1$ ($T_{\text{nat_le_Tchd}}$), which prices the preallocation of §7.1. - Theorem 3 extends the bound beyond $F(n)$ by the (A_M) case analysis above. That extension is not formalized.

Gate-C regime. At $d = m/n = \lg^\alpha n$ with $1/2 \leq \alpha < 1$ (bound expressions; only upper bounds on running times are claimed): - C-HD's certified bound is $\text{Tchd}(n, m) = \Theta(n \lg^{\alpha+1/3} n + n \lg^\alpha n \cdot \lg \lg n) = o(n \lg n)$. - DMSY26's bound is $\Theta(n \lg^{\alpha+1/2} n) \geq n \lg n$, and Dijkstra's bound is $\Theta(n \lg n)$.

7. Space

7.1 Formal (what is proved)

- In the model, $\text{space} \leq \text{cost}$ for every run (RAMAudit.run_space_le_cost). So the formalized program uses space $O(\text{Tdisp_F}(n, m))$.
- The formalized implementation (agent-09's dProSpecC; agent-04's DLayer instance) allocates the DS' arrays once, before the top call, and never re-allocates them: every D operation is allocation-free. It never reclaims stale entries.
 - Capacities:
 - * use capacity $\text{ucap} = 12 \cdot \text{kmN} \cdot T_{\text{nat}} + 2$, with $T_{\text{nat}} = \text{cn} + \text{cm} \cdot (tF + \lg(\text{dd}+1) + 3)$ and $\text{kmN} = \text{kmaster}(1, 1, 1000)$;
 - * entry pool, block records and block stack: $\text{ucap} + 2$ cells each;
 - * level tables $LF + 2$ cells, key lists $n + LF + 2$, live map n , select scratch $6 \cdot \text{ecap} + 60$.

- $ucap$ suffices because the spine threads the budget $use + 2 \cdot \lg \text{cost} \leq ucap$ (CallSpec), and $\lg \text{cost} \leq k_{\text{master}} \cdot T_{\text{chd}} \leq 6 \cdot k_{\text{master}} \cdot T_{\text{nat}}$ at the top.
- The allocation costs $O(T_{\text{nat}}) = O(T_{\text{chd}})$. RAM space is at most RAM cost, so it is within the time bound.
- Hence on the C-HD branch the space of the formalized program is $\Theta(T_{\text{nat}}) = \Theta(n + m \cdot (t + \lg d))$. The upper bound is formal (space $\leq$ cost); the lower bound is read off the program text (dPro allocates arrays of $ucap + 2$ cells) and is not a theorem. This is superlinear whenever $t = (\lg n/d)^{2/3} \rightarrow \infty$, i.e. in the whole Gate-C window. The formal Gate-C statement does not constrain space.

7.2 Sketch, NOT formalized: an $O(n + m)$ -space variant (inherits B1 S4.10/S5 §5)

- A reallocating implementation would reclaim stale entries and blocks when structures are destroyed.
- Graph, labels, versions, stamps and pointers: $O(n + m)$.
- Live D's: at most one per call on the recursion stack, $O(N_X)$ each, geometric in the level. Plus stale entries, bounded by the inserts since the structure's creation. $O(n + m)$.
- FindPivots state: $O(k)$ plus trees. Trees are vertex-disjoint and freed at return.
- Total $O(n + m)$. This variant is not the formalized program.

8. Novelty (agent-03 search log, 2026-09-20)

Searched sources: - the four supplied primary papers (2504.17033v2 DMM+25, 2602.07868v2 DMSY26, 2609.15247 KR26, 2609.04825v1 Cai); - B1 draft §8 novelty log (Semantic Scholar citers of DMSY26: KR26, Cai, 2609.14126, 2607.19346); - a web search (US index) for directed comparison-addition SSSP beating Dijkstra at moderate density: no hit beyond DMM+25/DMSY26. Findings. - DMSY26 Thm 1.1 gives $O(m \sqrt{\lg n} + \sqrt{(mn \lg n \lg \lg n)})$. This bound is $\Theta(n d \sqrt{\lg n}) \geq n \lg n$ for $d \geq \sqrt{\lg n}$. Its footnote 1 says the algorithm “also works when $\delta > \log k$ ”, but it analyses FindPivots as $k(\delta + \log k)$ per search. At $\delta = d$ that gives $\Sigma |Q| \cdot k \cdot d \geq m k d$, and the per-level scans give $n' L \delta = m L$. Neither yields $o(n \lg n)$ for $d \geq \sqrt{\lg n}$ (agent-03 notes §1: the balance pivots-failed $\geq m^2 \lg n$ in DMSY26's accounting). So C-HD is not implied by DMSY26 as written. - NEW elements: - (i) invalid targets become counted, unexplored LEAVES (agent-09), which makes a local search $O(k^2)$ independent of degree; - (ii) permanent deletion of edges into targets below L_X (agents 03/08/09/10; Lemma L); - (iii) the per-layer foreign-leaf bound via disjoint handled ranges, and home charging for contact-through-leaf re-selections (agent-03; §5.1-5.3); - (iv) the block-linking DS' Merge, whose total volume is independent of δ (agent-09; also agents 02/03/10); - (v) the parameter regime $k = \sqrt{t}$, $t = (\lg n/d)^{2/3}$. - KR26 Sec. 4 lists the directed extension of the undirected density bound as OPEN, via directed BundleDijkstra. C-HD does not solve that problem. It gives a weaker but NEW directed density improvement by a different mechanism. - Undirected $O(\sqrt{(mn \lg n)})$ (DMSY23/Yan26, KR26) is not comparable: undirected graphs only. - Independent logs: agent-09 /research/agents/agent-09/work/NOVELTY_LOG_CHD.md (web queries and hits checked) and agent-07's cross-check (board 11:23). Neither found a directed CA bound $o(n \lg n)$ at $d \geq c \cdot \sqrt{\lg n}$. - Dated re-run (v1.7, ~22:55): a new primary source was found, 2607.19346 (Hair-Li-Li-Zhang, July 2026). It is a Las Vegas randomized $m^{1+o(1)}$ algorithm for real, possibly negative, weights. Reviewer #2 (23:00) notes that it uses real subtraction (potentials), so it lies outside the comparison-addition model. It does not pre-empt C-HD. Its published analysis carries a factor $2^{\Omega(\sqrt{\lg n})}$, from a base case $\tilde{O}(m \cdot k)$ with k up to $2^{\sqrt{\lg n}}$. The factor is at most $\exp(O(\sqrt{\lg n} \lg \lg n))$, from at most $\sqrt{\lg n}$ recursion levels with $O(\log n)^i$ instances at level i . So the best bound known from it is $\omega(n \lg n)$ throughout the window (non-pre-emption confirmed independently by agents 02, 04, 05, 07, 09 and 10). Details are in the novelty log. - Dated re-run (v1.5): /research/agents/agent-03/work/paper/NOVELTY_LOG_2026-09-20.md (seven web queries, the four primary sources re-read, Cai's

Theorem 9.2 checked: worst case $\Omega(m \sqrt{\lg m})$ at $d \geq \sqrt{\lg n}$. No conflicting result found. Repeat immediately before any final claim. B1's Cor. S4-G is only $o(\text{DMSY26})$ for $m/n = o(\sqrt{\lg \lg n})$, so it does not overlap.

9. Formal verification and honest limitations (v2.0, 2026-09-20 ~23:15 UTC)

9.1 The formal chain (frozen build #33; all modules std axioms, no sorry, no project axiom)

- Top: $\text{Final.chd_gateC} : \text{Frontier.GateC} \Leftarrow \text{chd_CHDTarget} : \text{CHDTarget} \Leftarrow \text{chd_exact_within}$.
 - chd_exact_within holds for the closed program chdProgram , via $\text{L6/FinalGen.chdProg_of_DLayer}$, from concrete instances only.
 - All name side conditions are decided by decide_+kernel .
- Exactness and cost: $\text{MasterSpec} (\text{L6.masterSpec_of_goodP}, \text{GoodP} \text{ by } \text{goodP_chdP}; \text{OBJECTION 4 resolved through } \text{fpA_le_of})$ and the RAM refinement (the spine).
- The spine is a bounded level induction, agent-10's $\text{L6.LevelRoute.callSpec_of_levels} (\forall 1 \leq \text{LF}, \text{from level } 0 \text{ and the step } l \rightarrow l+1 \text{ for } l < \text{LF}), \text{ over agent-08's contract } \text{RamSpine.CallSpec} (\text{SpineLoop.lean})$. It uses ONE constant K at every level, and the D-layer use grows by at most 2 per unit of Layer-A cost. (agent-08's unbounded $\text{SpineLevels.callSpec_all}$ is not on the route.)
 - Base: agent-03's callSpec_zero . It covers the heap base case, the pointer pass, the level-0 DS' conversion and the heap clear.
 - Step: agent-01's $\text{body_spec} / \text{callSpec_succ}$, composed of:
 - * agent-09's FindPivots : fpB_inst , transported to the raw text $\text{fpAtRaw} (\text{BL2AtRaw}, \text{fpAt_eq_raw})$ and supplied as $\text{L6.hFP_of} (\text{BL2Step})$;
 - * agent-01's prologue (pro_spec);
 - * agent-08's main loop: the step's loop input is $\text{L6.hLoop_of_parts} (\text{SpineLoopInst})$, i.e. loop2_spec with agent-06's postSpec through $\text{RamSpine.loopSpecB_of_loop2} / \text{postHalf_cmpTT}$;
 - * agent-03's finalization ($\text{fin_spec}, \text{fin_spec_stmt}$).
 - The D layer: agent-04's DLayerF.DLf , with agent-02's merge and agent-05's pull.
 - The top: agent-10's dispatcher, L6 preprocessing, prologue, allocations (agent-09's dProSpecC) and output.
- The degenerate case $H.m = 0$ is a runtime branch $gM = 0$ of $\text{coreTop} (\text{complete_init_of_m0})$.
- Extras, corollaries of the frozen Final and outside its import closure (std axioms; verified by agent-10 at 23:03 UTC, and FinalCompare by agent-03 at 23:25). They enter the root at the post-audit build #34:
 - $\text{FinalExtras.chd_space}$ (agent-01) and $\text{FinalSpace.chd_space_within}$ (agent-05): cost and space $\leq C \cdot \text{Tdisp_F}$;
 - $\text{FinalExtras.chd_auditGateC}$ (Corollary 2'), via agent-09's $\text{AuditGateC.chdTarget_imp_auditGateC}$;
 - $\text{FinalCompare.chd_beats_known}$ (agent-01, 23:13): $o(\cdot)$ of the Dijkstra, DMM+25 and DMSY26 bound expressions along μ (§1);
 - FinalRaw (agent-09, 2d53cb7d): chdProgramRaw , the program as one literal term, and $\text{chdProgramRaw_eq} : \text{Final.chdProgram} = \text{chdProgramRaw} := \text{rfl}$. Both are axiom-free.
- Paper-to-code: `/research/final/THEOREM_MAP.md`, assembled by agents 07/09 from the owners' rows (agent-03's are in `PAPER_TO_CODE_agent03.md`).

9.2 Limitations (stated so that nothing is overclaimed)

1. **Formal density range.** The kernel-checked bound is $C \cdot \text{Tchd}$ only for $m \leq n \cdot F(n)$ with $F(n) = \lfloor \log_2 n \rfloor^{\{3/4\}}$. Outside that range, the formalized program uses Bellman-Ford, with the formal bound $C \cdot (n+1)(m+1)$.
 - The Gate-C statement is along $\mu(n) = n \cdot \lfloor \ln(n+2)^{\{3/4\}} \rfloor$, which lies inside the range. The strong form

- (Corollary 2') covers every graph with $m \leq n \cdot \max(1, F(n))$, and nothing denser.
- The paper's claim for $F(n) < d = o(\lg n / \lg \lg n)$ (Theorem 3) is NOT formalized.
2. **Space.** Formally, $\text{space} \leq \text{time}$: the implementation preallocates $\Theta(T_{\text{nat}})$ cells. The $O(n + m)$ -space variant (§7.2) is a sketch.
 3. **Constants.** The constant $C = \text{bodyC} \cdot K_{\text{cC}} + 65536 \cdot 9 + 100 = 2 \cdot K_{\text{cC}} + 1500 + 65536 \cdot 9 + 100 \approx 1.96 \cdot 10^{24}$ is astronomically large ($K_{\text{cC}} \approx 9.80 \cdot 10^{23}$; $dKd(12 \cdot \text{kmN}) = 23 \cdot 12 \cdot \text{kmN} + 300 \approx 2.82 \cdot 10^{12}$; agrees with REVIEW_1 §9.5).
 - $K_{\text{cC}} = (KC + 13000 + 34 \cdot (100 + dKd(12 \cdot \text{kmN}))) \cdot (\text{kmN} + 1) + 250$ (Final. K_{cC}), where $KC = 210000$ is the spine's one-per-level constant K (RAM cost per unit of Layer-A cost).
 - $\text{kmN} = \text{kmaster}(1, 1, 1000) = 5100 \cdot 35784 \cdot 56 \approx 1.02 \cdot 10^{10}$ is the Layer-A master constant ($L6/\text{DCap} \cdot \text{kmN}$; $\text{kmN}_{\text{le}} \leq 10^{11}$).
 - The result is purely asymptotic and makes no practical claim.
 4. **Word size.** The machine's word cap is $(n+m+2)^{e+1}$ with $e = 160$, i.e. $O(\log n)$ -bit words (e is a constant). Reals are only compared and added.
 5. **Determinism and model.** The program is deterministic and works in the comparison-addition model. It is compared with the best known directed bounds in that model. 2607.19346 does not compete in the window (§8): it is Las Vegas randomized, runs in $m^{1+o(1)}$ with an implicit $m^{o(1)}$, and uses real subtraction (potentials), so it is outside comparison-addition; the last point is per reviewer #2.
 6. **Reviews and reproduction** (state at 00:00 UTC, 2026-09-21).
 - agent-09's final audit and kernel replay of frozen #33 passed at 23:37: 18,994 constants, 266 modules, std axioms.
 - agent-07's fresh-copy rebuild passed (2548 jobs, 0 HARD tokens), with an own kernel replay of 18,994 constants.
 - REVIEW_1 (agent-07, 23:57): ACCEPTED for Gate (C), with limitations.
 - agent-02's fresh-copy rebuild and REVIEW_2 are pending.
 - agent-10 integrates the final/ files only after all of these pass.
 7. **Novelty.** See the dated log NOVELTY_LOG_2026-09-20.md, last re-run ~22:55 UTC (§8). The web index is US-only and may lag arXiv.
 8. **Size of the gain.** At $d = m/n = \lg^\alpha n$ ($1/2 \leq \alpha < 1$), C-HD's certified bound is below Dijkstra's $\Theta(n \lg n)$ bound by the factor $\lg^{\{(1-\alpha)/3\}} n$ (§6). Only upper bounds are proved; no lower bound on C-HD's running time is claimed. That is $\lg^{1/12} n$ along the formal profile μ ($\alpha = 3/4$), and at most $\lg^{1/6} n$ ($\alpha = 1/2$). It is a density-regime (Gate C) result, not a Gate A/B result.
 - On sparser graphs nothing new is claimed. For $\alpha \leq 1/4$, DMSY26's $O(m \sqrt{\lg n} + \sqrt{mn \lg n \lg \lg n}) = O(n \lg^{\{\alpha+1/2\}} n)$ is at least as small as C-HD's bound $T_{\text{chd}} = \Theta(n \lg^{\{(2+\alpha)/3\}} n)$ (for $\alpha > 0$). As bound expressions, C-HD's is smaller than DMSY26's exactly when $\alpha > 1/4$, and smaller than Dijkstra's when $\alpha < 1$.
 9. **Undirected graphs.** For undirected graphs, the known $O(\sqrt{mn \lg n})$ (DMSY23/Yan26; KR26) is stronger than T_{chd} throughout the window: $(1+\alpha)/2 < (2+\alpha)/3$ for $\alpha < 1$. C-HD's improvement is for directed graphs only.
 10. **Where the C-HD code runs.** The C-HD branch runs only for $\log_2 n \geq 16$ and $\lceil m/n \rceil \leq F(n)$. For $\log_2 n < 16$ the program runs Bellman-Ford, but the formal bound there is still $C \cdot T_{\text{chd}}$ whenever $m \leq n \cdot F(n)$, through the term 65536·9 of C (§6).